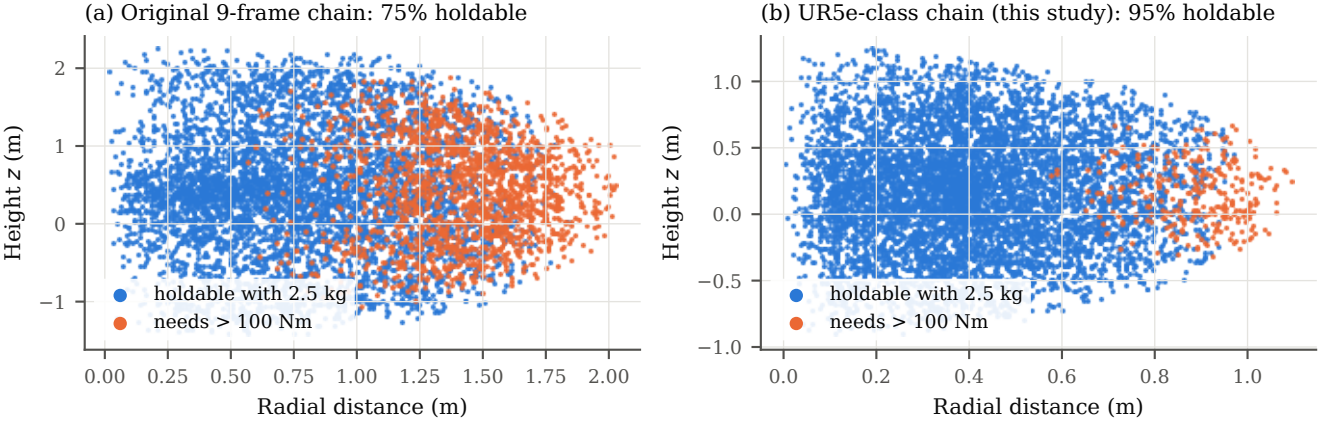

(a) Original 9-frame chain: 75% holdable
(b) UR5e-class chain (this study): 95% holdable
Height z (m)
Radial distance (m)
holdable with 2.5 kg
needs > 100 Nm
2
1
0
−1
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
1.0
0.5
0.0
−0.5
−1.0
0.0
0.2
0.4
0.6
0.8
1.0